



THE ROMAN CATHOLIC DIOCESE OF SUMBAWANGA

**FORM FOUR MOCK EXAMINATION FOR
DIOCESAN SECONDARY SCHOOLS 2026**



032/1

CHEMISTRY 1

Time 3:00 Hours

2026.

Instructions

1. This paper consists of three sections A, B and C with a total of **eleven (11)** questions.
2. Answer all questions in sections A and B and only **two (2)** questions for section C.
3. Section A carries **sixteen (16)** marks, section B carries **fifty-four (54)** marks and section C carries **thirty (30)** marks.
4. **Non-programmable** calculators may be used.
5. Cellular phones and any unauthorized materials are **not** allowed in the Examination Room.
6. Write your **Examination Number** on every page of your answer booklet(s).
7. The following constants may be used:
Atomic masses: H=1, C=12, N=14, O=16, Na=23, Cl=35.5, K=39, Ca=40, Mg=24, S=32,
Ag=108, Cu=63.5, Fe=56, Pb=208
 $1F = 96500 \text{ Coulombs/mol}$
 $1 \text{ litre} = 1 \text{ dm}^3 = 1000 \text{ cm}^3$
Molar volume of gases = $22.4 \text{ dm}^3/\text{mol}$

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Answer all questions in this section

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- i. What volume of hydrogen gas will be produced when 1.3 g of zinc granules react completely with excess dilute sulphuric acid at S.T.P?

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- D. 440 cm^3

E. 448 cm³

- #### A. Wet sand and carbon dioxide

B. Dry foam and foam

C. Dry powder and carbon dioxide

D. Sand and wet chemicals

E. Wet sand dry powder

- ### A. Gaining process

B. Reduction process

C. Oxidation reaction

D. Loosing process

E. Oxidizing reaction

- ### A. Data collection

B. Drawing conclusion

C. Data analysis

D. Experimentation

E. Data interpretation

- A. Spoon cathode and copper anode

B. Spoon as anode and copper as cathode

C. Spoon cathode and copper (ii) sulphate anode

D. Carbon cathode and copper anode

E. Spoon anode and carbon cathode

- $$\text{Fe}_2\text{O}_{3(s)} + 3\text{C}_{(g)} \rightarrow 2\text{Fe}_{(s)} + 3\text{CO}_{(g)}$$

Which one of the sentences is correctly describes this reaction?

A. Thermal decomposition

B. The carbon get reduced

C. The iron oxide gets reduced

D. The iron oxide gets oxidized

E. Carbon neutralizes the iron oxide

- ### A. Powder

B. Granules

C. Turning

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D. Ribbon

E. Marble chips

- viii. During qualitative analysis practical most of the student forget an apparatus used to scoop solid from one point to another especial from the watch glass to the test tube. What is the name of such apparatus?
- A. Spatula
B. Dropper
C. Test tube holder
D. Forceps and boss
E. Beaker and racks
- ix. Juma demonstrated an experiment by dissolving sugar in water until the solute was not dissolving any more. What type of solution was formed at the end of her demonstration?
- A. Saturated solution
B. Emulsion
C. Un saturated solution
D. Suspension
E. Super saturated solution
- x. A farmer wants to improve soil fertility and water retention in a sustainable way. Most of the ways used is destructive methods of soil conservation, which option is most suitable?
- A. Applying more chemical fertilizers
B. Using only only pesticides
C. Adding well-decomposed manure
D. Burning crop residues
E. Irrigating more frequently
2. Match the descriptions in List A with the corresponding characteristics of living things in List B, by writing the correct match against the items number in your answer sheet provide.

List A	List A
i. Plant nutrients are washed into the lower layer of the soil	A. Leaching
ii. The nutrient content of manure from the animals	B. Green manure
iii. The capacity of the soil to hold and allow important nutrients flow	C. Strait fertilizer
iv. Trace elements	D. Farmyard
v. Contain two or more macronutrients	E. Soil fertility
vi. Covering soil with grasses to prevent loss of water	F. Micronutrients
	G. Compound fertilizer
	H. Soil productivity
	I. Mulching

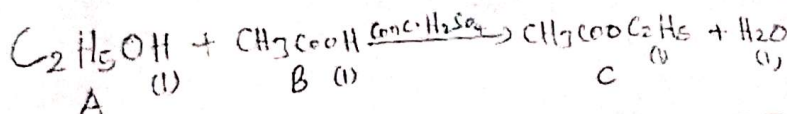
SECTION B (54 Marks)

Answer all questions in this section

3. There are many scientists who explained about an atoms based on structure and the arrangements of electrons in the orbits. What were the contribution of Rutherford on atom? Give six points.
4. Justify the following statements as used in chemistry:
- a) Chemists prefer to use Bunsen burner as the source of heat
b) Carbon is a non-metal
c) Concentrated sulphuric acid is not an electrolyte
d) Melting of ice is regarded as physical change
e) In filtration, the filtrate passes through the filter paper while residue does not
f) It is advised to use evaporating dish instead of conical flasks to evaporate a solution
5. A certain compound has the following composition, 20.10% iron, 11.50% sulphur, 23% oxygen and

the rest is water of crystallization. If the vapour density of the compound is 139

- Calculate the molecular formula of the compound
 - Write the name of the compound formed
 - Write the uses of the compound formed (three points).
 - How cation in the compound formed being useful in daily life
6. Three moles of nitrogen gas combined with five moles of hydrogen gas to form ammonium gas by Haber process. This method is commonly used in industrial production of ammonia, by considering that reaction between that reactants attempt the question below.
- Which reactant is present in small amount?
 - Calculate the gram of reactant left in the container?
 - How many moles of ammonia are produced at s.t.p?
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7. When dilute nitric acid is added to green solid P a blue solid X is formed with evolution of a gas R that forms a white precipitate with lime water. Heating the blue solution of the hydrated salt X, it initially loses water then decomposes to form black solid S, brown fumes of gas Y and gas Z.
- Identify the solid X, gas Y and gas Z
 - Write an equation for the reaction between solid P and dilute nitric acid
 - Write an equation for formation of solid S
 - Explain two uses of gas R in daily life
8. Two organic compounds react together as summarized by equation shown below.



- Name the compounds marked A, B and C.
- What is the role of concentrated sulphuric acid in the reaction?
- Name the homologous series to which compounds A and B belongs.
- State two uses of compound C
- What type of chemical reaction is portrayed by the equation?

SECTION C (30 Marks)

Attempt two (2) questions in this section

9. Assume that you are chemist in a chemical plant that deals with the production of chlorine gas. If you want to produce 100 litres of chlorine gas per hour so that you can reach the company's goal of producing 2400 litres every day. What current of electricity will you allow to flow per hour?
10. A piece of land requires 120Kg of N, 60 Kg of P_2O_5 , and 80kg of K_2O to be applied per hectare. The following fertilizers are available for application:
Sulphate of ammonium, SA (21% N), single superphosphate, SSP (18 % P_2O_5) and Nitrate of potash, NOP (60 % K_2O). Calculate the amount of each fertilizers needed to be applied per hectare on the piece of land.
11. In six points explain how changes of states of matter are significant in the life of human being.